

quality and quantity of data directly impact the performance of ML models, highlighting the importance of collecting, cleaning, and preparing data for effective machine learning applications.

## Types of machine learning

### Supervised learning

Supervised learning involves training a model on a labelled dataset, where the algorithm learns the relationship between input features and corresponding output labels. Common examples include:

- *Image classification:* Identifying objects in images.
- *Spam detection:* Differentiating between spam and non-spam emails.

**How it works:** During training, the algorithm adjusts its parameters to minimise the difference between predicted and actual labels. The trained model can then make predictions on new, unseen data.

### Unsupervised learning

Unsupervised learning deals with unlabelled data, aiming to discover patterns or relationships without predefined outputs. Examples include:

- *Clustering:* Grouping similar data points together.
- *Dimensionality reduction:* Simplifying complex datasets while retaining essential information.

**How it works:** Algorithms in unsupervised learning explore the inherent structure within the data, revealing hidden patterns or relationships without explicit guidance.

### Reinforcement learning

Reinforcement learning involves an agent learning through trial and error, receiving feedback in the form of rewards or penalties. Examples include:

- *Game-playing AI:* Learning optimal strategies by playing games.
- *Robotic control:* Teaching robots to perform tasks through continuous feedback.

**How it works:** The agent takes actions in an environment, receives feedback, and adjusts its strategy to maximise cumulative rewards over time.

## Key concepts in machine learning

**Features and labels:** In the context of machine learning, features are the measurable properties or characteristics of the data. For example, in an image recognition task, features might include pixel values, colours, and shapes. Choosing relevant features is crucial as they directly impact the model's ability to learn and make accurate predictions.

**Explanation of labels/targets:** Labels, also known as targets or outputs, represent the desired outcome of the learning process. In a supervised learning scenario, the

algorithm aims to predict these labels based on input features. For instance, in a spam detection model, the labels would indicate whether an email is spam or not.

### Training data and testing data

**Importance of splitting data:** To evaluate the performance of a machine learning model, it's essential to split the available data into two sets — training data and testing data. The model learns patterns from the training data, and the testing data assesses its ability to generalise to new, unseen examples. This division ensures a fair evaluation of the model's performance.

**Overfitting and underfitting:** Overfitting and underfitting are common challenges in ML.

- Overfitting occurs when a model learns the training data too well, including noise or irrelevant patterns, resulting in poor performance on new data.
- Underfitting happens when a model is too simplistic and fails to capture the underlying patterns in the data, leading to poor performance overall.

Balancing between overfitting and underfitting is a crucial aspect of building effective ML models.

## Common ML algorithms

### Linear regression

**Concept and use cases:** Linear regression is a foundational supervised learning algorithm used for predicting a continuous outcome based on one or more input features. The model assumes a linear relationship between the input features and the target variable. Use cases include:

- *Predicting house prices:* This is based on features like square footage, number of bedrooms, and location.
- *Revenue prediction:* This involves estimating sales revenue based on marketing expenses.

**How it works:** Linear regression aims to find the best-fitting line (linear equation) that minimises the difference between predicted and actual values. This line is determined by adjusting weights assigned to each input feature during the training process.

### Decision trees

**Concept and use cases:** Decision trees are versatile algorithms used for both classification and regression tasks. They represent decisions as a tree structure, with each node representing a decision based on a specific feature. Use cases include:

- *Credit scoring:* Determining creditworthiness based on factors like income, debt, and credit history.
- *Medical diagnosis:* Classifying diseases based on patient symptoms.

**How it works:** Decision trees split the dataset based on features to create subsets that are as pure as possible with